MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS8NW6 (INCLUDING VARIATION LS8NWA6)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24643.

This cable can be manufactured in two variations: LS8NW6 (unarmored) and LS8NWA6 (armored).

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor, uncoated. AWG size 12, class B stranding.
Second - Separator (white) may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Cross-linked polyethylene or ethylene propylene rubber insulation, colored black. Nominal thickness 0.030 inch. Standard identification code applied by method 1.
Fourth - Eight conductors cabled together with a lay not greater than 24 times the pitch diameter. Fillers shall be employed as necessary to form a firm, well-rounded assembly.
Fifth - Binder tape applied helically with overlap.

Construction variant for type LS8NW6

Sixth - Cross-linked polyolefin jacket. Nominal thickness of 0.040 inch. Maximum overall diameter 0.670 inch. Cable surface marking required.

(B) denotes changes.

AMSC N/A FSC 6145
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Construction variant for type LS8NWA6

(B) Sixth - Cross-linked polyolefin jacket. Nominal thickness of 0.040 inch. Maximum overall diameter 0.670 inch. Cable surface marking not required.
Seventh - Braided metal armor and paint.

NOTES: 1. Variant LS8NW6 military part number - M24643/54-01UN.
2. Variant LS8NWA6 military part number - M24643/54-02AN.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum.	1.72
Voltage withstand - conductor to conductor, volts, root mean square, minimum.....	4000
Insulation resistance - megohms/1000 feet, minimum.....	4200
Conductor continuity.....	No failure
Group A:	
Visual and dimensional.....	No failure
Group B:	
Cold bending, cable - 5-inch mandrel diameter, maximum..	No damage
Drip - 95 ± 1°C.....	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)..	35
Physicals (unaged)	
Insulation	
Cross-linked polyethylene	
Tensile strength - lbf/in^2, minimum.....	1800
Elongation - percent, minimum.....	250
Ethylene propylene rubber	
Tensile strength - lb/in, minimum.....	700
Elongation - percent, minimum.....	250
Jacket (cable)	
Tensile strength - lbf/in^2, minimum.....	1300
Elongation - percent, minimum.....	160
Group C:	
Physicals (aged)	
Insulation	
Cross-linked polyethylene	
Air oven	
Tensile strength - percent of unaged, minimum.....	80
Elongation - percent of unaged, minimum.....	80
Ethylene propylene rubber	
Air oven	
Tensile strength - percent of unaged, minimum.....	75
Elongation - percent of unaged, minimum.....	75

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EXAMINATION AND TESTS: (Continued)

	Requirements
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum	50
Elongation - percent, minimum	50
Shrinkage	No failure
Permanence of printing (conductor) - cycles, minimum	25
Permanence of printing, type LS8NW6 only (jacket) - cycles, minimum	250
Cable filler removability	No failure
Heat distortion - percent of unaged, maximum	30
Armor - conformance to materials, construction and coverage (type LS8NWA6 only)	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

	Cold working (minus 20 ± 2°C)	No damage
	Aging and compatibility (cable) (125 ± 5°C)	No failure
(B)	Abrasion resistance (jacket) - scrapes, minimum	75
	Acid gas equivalent - percent, maximum	
	Jacket	2
	Fillers	2
	Insulation	18
	Halogen content - percent, maximum	
	Jacket	0.2
	Fillers	0.2
	Flame propagation (cable)	No failure
	Immersion (jacket)	
	Tensile strength - percent of unaged, minimum	50
	Elongation - percent of unaged, minimum	50
	Smoke index, maximum	
	Jacket	25
	Fillers	--
(B)	Insulation	45
	Toxicity index, maximum	
	Jacket	5
	Fillers	5
	Insulation	1.5
	Durometer (jacket) - (type A) hardness, minimum	80
	Weathering (jacket)	No failure

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UNIT ORDERING LENGTH: 2000 feet (nominal)

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

Preparing activity:
Navy - SH
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